INCIDENT LIFE CYCLE AUTOMATION

Functional System Design (FSD) Documentation

Implementation-oriented design for Incident Life Cycle Automation

STUDENT NAME Nareshkumar S	PROJECT **Incident Life Cycle Automation**
DOCUMENT DATE **08 October 2026**	SCOPE **Incident lifecycle workflow & automation only**

1. Introduction

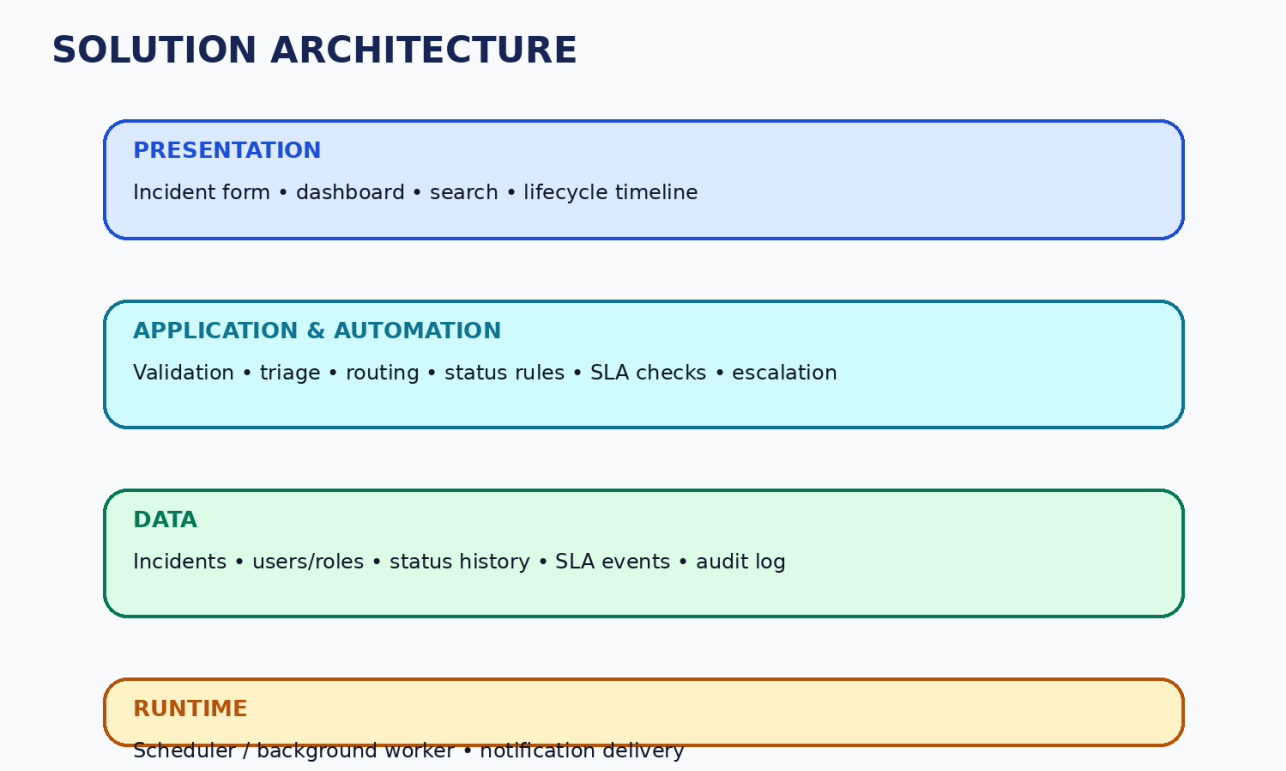
Project Title: Incident Life Cycle Automation
Student Name: Nareshkumar S

2. Project Overview

The system manages operational incidents as a continuous workflow from intake to closure. The design focuses on visibility, ownership, SLA control and traceable history.

- Incident intake and unique identification
- Classification and priority
- Assignment and reassignment
- SLA monitoring and escalation
- Lifecycle status and work notes
- Resolution and closure
- Search, dashboard and audit history

3. Architecture



4. Logical Data Model

Entity	Key Data	Relationship
Incident	ID, title, description, source, category, impact, urgency, priority, status, owner, timestamps	Central record linked to users, history, SLA events and resolution.
User / Role	User ID, name, role, access scope	Controls who can view or modify incidents.
Status History	Incident ID, old/new status, actor, timestamp, note	One-to-many relationship with Incident.
SLA Event	Incident ID, due time, event type, triggered time, state	Tracks SLA and escalation lifecycle.
Audit Entry	Actor, action, entity, timestamp, context	Provides traceability for key changes.

5. Service Responsibilities

Service	Responsibility
Incident Service	Create, retrieve, update and close incidents.

Triage Service	Apply classification and priority rules.
Assignment Service	Resolve queue/owner and handle reassignment.
SLA Worker	Check due times and trigger escalation events.
Notification Service	Deliver assignment, status and SLA messages.
Audit Service	Store lifecycle and administrative history.

6. Testing

Functional, UAT and performance testing should be performed against the actual implementation. Results should be recorded from execution rather than assumed.

7. Known Limitations

The initial design focuses only on core incident lifecycle automation. Advanced integrations and analytics are not required for the baseline project.

8. Future Enhancements

Additional service-specific SLA policies, deeper reporting and selected external integrations can be considered after the core lifecycle is stable.